Reflection and contribution journal

Project: Telco Customer Churn - EDA and preprocessing
Group members: Viviana Zamora, Patrick Ray, Dustin Mastrota, Funsho Orogun
Course / section: ITAI-1374
Instructor approval: September 27, 2026, as reported by Viviana.
Execution date: September 20, 2026

1. Objective and scope

The project prepares an IBM educational customer sample for a future binary classification task. The target is whether a customer churned. This midterm covers EDA and preprocessing; no predictive model was trained and no model performance is reported. The original Kaggle dataset was preserved without manual edits.

2. Verified processing results

Measure	Observed result
Original dataset	7,043 rows, 21 columns
Raw training partition	4,930 rows
Untouched testing partition	2,113 rows
Exact duplicate training records	0
Repeated profiles excluding customer ID	11 (retained as distinct customers)
Missing TotalCharges in training	8
Training TotalCharges median used for filling	1,415.28
Original training labels	No: 3,622; Yes: 1,308
Training churn rate	26.53%
Numeric predictor columns	49 (plus separate ID and target)
One-hot indicator columns	41
Separate balanced training copy	7,244 rows; 3,622 in each class
Additional sampled copies	2,314

The 70/30 allocation is approximate at the individual-row level because 30% of 7,043 is not an integer. The split uses seed 42 and stratifies on the target. No customer ID appears in both partitions. The holdout was saved as original raw records and removed from the analysis workspace. Its file hash remained unchanged through preprocessing.

All EDA summaries and plots use training data. Text in numeric columns is converted in code, missing numeric values are filled with training medians, and an indicator records missing TotalCharges. Meaningful service categories such as `No internet service` are retained. Numeric ranges are checked; large but plausible charges are not automatically deleted.

Feature engineering adds internet subscription status, service count, and a fixed flag for tenure of 12 months or less. Customer IDs are metadata rather than predictors. Churn is encoded as 0/1 in the numeric exports; categorical predictors are one-hot encoded. Training means and standard deviations determine scaling, and training categories determine encoding. TotalCharges also receives a fixed square-root transformation. A separate oversampled training copy demonstrates balancing while the main exports retain natural class proportions.

3. What the plots show

Missing-value plots document the training values filled by the imputer. The before/after numeric plots distinguish changes in units from changes in distribution shape: standard scaling changes location and scale, whereas the power transformation can change skewness. On the same imputed training rows, TotalCharges sample skewness changed from **0.954** to **0.306**. This does not prove normality or better model performance.

The contract plot shows the following training-only association:

Contract	Training customers	Observed churn rate
Month-to-month	2,708	42.69%
One year	1,013	11.94%
Two year	1,209	2.56%

These associations motivate future predictor comparisons. They do not establish a causal effect of contract type. The class-balance plots show the original imbalance and the equal class counts produced by random oversampling. Sampling with replacement repeats existing customers and adds no new information.

4. Limitations and alternatives

• Median filling is an approximation. The missingness flag preserves information about absence, but a different imputation rule could be compared in training validation.

• A transformed numeric distribution may help some models and be unnecessary for tree-based models. Transformations should be evaluated with the eventual classifier.

• Engineered features overlap with original columns and may add little value. Compare versions within training folds.

• Oversampling can overfit repeated observations. Compare it with natural class prevalence and class weights.

• The sample is historical and educational. Define the prediction time and verify feature availability before claiming a prospective business use.

5. Verification and future work

The notebook ran all cells and checked source and holdout hashes, disjoint split IDs, output dimensions, numeric completeness, and balanced training counts. All preparation checks passed. The emailed file `06_cleaned_training_data.csv` contains 4,930 rows and 25 columns. The numeric file contains 49 predictors, plus customerID and Churn. The notebook records detailed results and recreates `reports/metrics.json` when run.

For the final project, begin cross-validation from `data/splits/train_raw.csv`. Refit imputation, encoding, scaling, and any balancing within each training fold. Do not cross-validate the already oversampled or globally preprocessed exports. Compare baseline, logistic regression, and tree-based classifiers using precision, recall, F1, ROC-AUC, and precision-recall measures. Select the workflow using training validation, then evaluate once on the holdout at its natural class proportions.

Team reflection

These entries are intentionally blank. Record actual experiences and decisions; the prepared technical record does not establish any member's personal contribution.

Instructor approval date and evidence

Instructor approved September 27, 2026; date reported by Viviana.

A decision we discussed: alternatives and the reason for our choice

We considered removing rows with missing TotalCharges values or filling those values. We chose the training median because it preserves customer records and is less affected by unusually high charges than the mean. We calculated the median using only training data to keep the testing data untouched.

A difficulty we encountered and how we resolved it

Our main difficulty was coordinating the project within the available time. Waiting for the instructor’s response about dataset approval added time pressure, and organizing responsibilities around different schedules made group discussions challenging. We addressed these difficulties by keeping communication open through GroupMe, which allowed us to share updates and coordinate tasks even when we could not all meet at the same time.

What we learned and can explain

We learned that a realistic project timeline needs to include time for instructor approval, feedback, and coordinating everyone’s schedules. We also learned the importance of clear responsibilities and regular updates. Using GroupMe helped us keep communication open and coordinate tasks even when our schedules did not allow us to meet at the same time.

What we changed after reviewing the draft and how we checked it

After reviewing the draft, we corrected the instructor approval date to September 27, 2026. We also revised the reflection to describe the project’s actual challenges: waiting for approval, organizing responsibilities, and coordinating different schedules. We clarified that we addressed these challenges by keeping communication open through GroupMe. Our review focused on making the dates and descriptions accurate and consistent with our experience.

What we would improve next, based on the results

For future projects, we would set earlier internal deadlines, assign clear responsibilities from the beginning, and use regular GroupMe check-ins to track progress. We would also allow more time for instructor feedback and a final group review of the notebook, data, and documents. These changes would help distribute the work more evenly and identify unfinished tasks before the submission deadline.

How the team reviewed and used assistance

We reviewed one another’s contributions and checked the project materials for accuracy, clarity, and consistency with the assignment requirements. During our review, we corrected the instructor approval date and revised the reflection to accurately describe our team’s challenges, communication through GroupMe, and plans for improvement.

Individual contribution journal

Each member completes their own entry using only work actually performed.

Viviana Zamora

Dates worked and date this entry was reviewed

09/20-30/2026

Work personally completed and decisions or problems addressed

I coordinated the preparation and organization of our Telco Customer Churn project files . I created PDF versions of the written documents and kept the Jupyter notebook in `.ipynb` format so Patrick could open and review it. I worked through file-opening difficulties and confirmed that our professor approved the dataset on September 27, 2026. I also coordinated with Patrick about sending the files by email; he is responsible for the final group submission.

Evidence: notebook sections, files, figures, checks, or commit links

Files related to my contribution include 02_project_proposal.pdf, 03_eda_preprocessing.ipynb, and 05_opening_and_submission_guide.pdf. These are examples of the project documents, notebook, and instructions whose preparation and file formats I helped coordinate.

What I learned and can explain

I learned that different project files require different programs: PDFs are for reading documents, Jupyter opens .ipynb notebooks, and CSV files contain the dataset. I also learned that a file failing to open does not necessarily mean it is corrupted—it may be opening in the wrong program. Clear filenames and opening instructions help group members review and use the project files.

Patrick Ray

Dates worked and date this entry was reviewed

Work personally completed and decisions or problems addressed

Evidence: notebook sections, files, figures, checks, or commit links

What I learned and can explain

Individual contribution journal

Dustin Mastrota

Dates worked and date this entry was reviewed

Work personally completed and decisions or problems addressed

Evidence: notebook sections, files, figures, checks, or commit links

What I learned and can explain

Funsho Orogun

Dates worked and date this entry was reviewed

Work personally completed and decisions or problems addressed

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What I learned and can explain